# Cluster Computing Using ScaleMP: A Comparative Analysis of Virtual SMP and Distributed Memory Architectures

Syarifah Dania Binti Syed Abu Bakar

*Universiti Teknologi Malaysia*

syarifahdania@graduate.utm.my

***Abstract*****—** Cluster computing has become a fundamental technique to deal with the increasing demands of high performance data processing. Traditional distributed systems scale very well but typically add complexity since they are based on message passing and separate memory spaces. This paper discusses cluster computing with special emphasis on ScaleMP that uses a virtual symmetric multiprocessing (vSMP) architecture which enables multiple nodes to work as one shared memory system. This paper covers the architectural design and performance features of ScaleMP and compares it to traditional distributed clusters and parallel computing models. Shared-memory architectures can provide significant performance gains and simplified programming complexity for memory-intensive applications [1], [7]. However, distributed systems are still better in scalability for large scale workloads [6]. In summary, the paper shows the tradeoffs between scalability, performance and programmability and demonstrates that ScaleMP is a viable alternative for specific high-performance computing scenarios.

***Keywords*****—** Cluster Computing, ScaleMP, Virtual SMP, Distributed Systems, High-Performance Computing

## I. INTRODUCTION

The exponential growth of data generated by modern applications has brought significant challenges to the traditional computing systems. As cluster computing is rising in this modern era, single-node architectures are unable to efficiently process large-scale workloads. Cluster computing distributes computational tasks across multiple interconnected nodes to enhance processing speed, scalability, and fault tolerance.

The traditional cluster architectures are based on distributed memory models, where each node works independently and communicates via message passing. This approach allows high scalability, but also adds communication overhead and increases the system complexity [6].

To overcome these limitations, alternative architectures such as ScaleMP have been developed. ScaleMP has proposed a virtual symmetric multiprocessing (vSMP) model that clusters a set of nodes into a shared memory system, facilitating application design and minimizing the communication overhead [7].

In this paper we analyse the architecture and performance of ScaleMP and compare it to traditional distributed cluster systems and parallel computing models.

## II. TECHNOLOGY ARCHITECTURE

Cluster computing systems are made up of many interconnected nodes that work together to perform tasks in parallel. These systems are designed to increase computational efficiency and allow for large scale data processing.

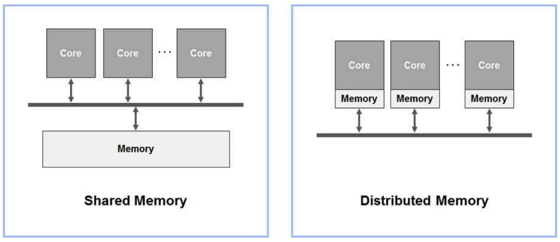


*Figure 1. Cluster vs ScaleMP Architecture*

The basic architectural differentiation between conventional distributed cluster systems and ScaleMP's virtual symmetric multiprocessing (vSMP) paradigm are shown in Figure 1. Each node in a distributed architecture keeps its own memory and uses message passing to interact, which raises programming complexity and latency. On the other hand, programs can access data without explicit communication thanks to the vSMP model's unified shared-memory space between nodes. Although preserving memory coherence between nodes becomes more difficult, this architectural distinction greatly lowers communication overhead and streamlines application development [1], [4].

ScaleMP, which implements a virtual SMP architecture, is a major breakthrough in this field. ScaleMP offers a shared-memory abstraction spanning several machines, in contrast to conventional distributed systems that depend on explicit communication between nodes. According to research, shared-memory systems can enhance in-memory application performance, especially for workloads based on OpenMP [1].

Virtualization approaches also facilitate system flexibility and effective resource usage [2]. However, compared to distributed architectures, preserving memory consistency between nodes adds overhead that may restrict scalability. Research shows that because distributed processing and storage systems may scale horizontally, they are still better suited for very big datasets [3].

## III. COMPARATIVE ANALYSIS

### i) Comparing ScaleMP with Distributed Cluster Systems

Conventional distributed clusters necessitate network-based message forwarding for communication and rely on distinct memory regions. High scalability is made possible by this, but it also adds latency and complicates programming [6].

ScaleMP, on the other hand, offers a single shared-memory environment that lowers communication overhead and streamlines development.

Table 1

| Criteria | Distributed Clusters | ScaleMP (vSMP) |
|---|---|---|
| Memory Model | Distributed | Shared (virtual) |
| Communication | Message passing | Implicit |
| Scalability | Very high | Moderate |
| Complexity | High | Lower |

*Table 1. Comparison between ScaleMP and Distributed Cluster.*

### ii) Comparing Parallel Processing Models (MPI/OpenMP) vs. ScaleMP

MPI and other parallel computing frameworks are powerful yet complicated because they require direct communication between nodes. Although OpenMP is usually restricted to single machines, it does offer shared-memory processing.

Large-scale shared-memory processing is made possible by ScaleMP, which expands OpenMP's capabilities over several nodes and enhances performance for workloads requiring a lot of memory [7]. However, because MPI-based systems have more precise control over communication, they might be more efficient in optimal situations.

### iii) Comparing ScaleMP and Conventional SMP systems

Conventional SMP systems have low latency but less powerful in terms of scalability since they rely on physically shared memory within a single computer or machine. On the other hand, by virtualizing shared memory across several nodes, ScaleMP allows greater memory capacity and enhanced speed which improve computing efficiency for scientific workloads [5].

## IV. CASE STUDY (NETFLIX)

In order to provide personalized recommendations to Netlix users, they run a big data infrastructure that analyzes large amounts of user interaction data. Netflix uses distributed cluster computing platforms that allow scalable data processing across several nodes in order to facilitate data analysis. With excellent scalability and performance, these technologies enable Netflix to handle millions of users simultaneously. However, data flow between nodes in distributed systems often resulting in communication overhead which can increase latency and system complexity [6]. ScaleMP offers a different strategy in this situation by providing a shared-memory environment across several nodes. This enhances performance for memory-intensive workloads and lessens the need for direct communication [1]. Although Netflix generally uses distributed systems for scalability, this case study depicts how shared-memory architecture like ScaleMP can be helpful for certain type of jobs that require quick data access with low latency.

## V. CONCLUSION

With an emphasis on ScaleMP, this work has investigated cluster computing. The analysis demonstrates that ScaleMP's shared-memory architecture offers notable benefits in terms of programmability and decreased communication overhead.

However, because of their higher scalability, distributed cluster systems continue to be more appropriate for large-scale applications. Hybrid architectures that combine the advantages of distributed and shared-memory systems may be investigated in future studies.

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**LOGBOOK**

| Date | Search Keywords | Databases / Tools Used | AI Prompts Used | Task Description |
|---|---|---|---|---|
| 20 April 2026 | “ScaleMP vSMP architecture”, “cluster computing shared memory vs distributed” | Google Scholar, IEEE Xplore | What is ScaleMP and how does it work | Started research on topic. Read abstracts of multiple papers and identified ScaleMP as main focus. Saved relevant articles. |
| 21 April 2026 | “ScaleMP performance OpenMP”, “shared memory cluster evaluation” | Google Scholar, Scopus | Explain difference between shared memory and distributed memory | Selected key papers for references. Took notes on ScaleMP architecture and advantages. |
| 22 April 2026 | “distributed systems vs shared memory HPC”, “MPI vs OpenMP comparison” | Google Scholar, Scopus | Help me understand MPI vs OpenMP | Analysed comparison between distributed clusters and shared memory systems. Drafted comparison ideas. |
| 23 April 2026 | “ScaleMP vSMP case study”, “HPC applications shared memory” | Google Scholar, Web of Science | How to structure academic paper introduction | Started writing introduction and technology section. Focused on explaining ScaleMP clearly. |
| 24 April 2026 | “cluster computing architecture diagram”, “virtual SMP diagram” | Google Images, draw.io | Where to place figure in IEEE paper | Created architecture diagram and added explanation. Continued writing main body. |
| 25 April 2026 | “ScaleMP vs distributed clusters comparison”, “OpenMP performance study” | Google Scholar, Scopus, ACM | Help me improve comparative analysis section | Completed comparative analysis section. Added table comparing ScaleMP and distributed systems. |

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|---|---|---|---|---|
| 26 April 2026 | “Netflix big data cluster computing case study” | Google Scholar, ACM, Scopus | Help me write case study | Wrote case study section and linked it to ScaleMP concept. Added citations. |
| 27 April 2026 | “IEEE paper format example”, “APA vs IEEE citation” | IEEE template, Word | Convert references into IEEE style | Finalized paper. Inserted references, formatted citations, checked grammar, and prepared for submission. |